
系统开发工具基础

实验报告

实验周次：第 1 周

学 院 信息科学与工程学部

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实验日期 2026.8.24

一、 课堂练习

1. 题目一：含空格文件名的批量整理

(1) 题目内容

题目要求

- 创建 input/docs 和 input/tmp；创建 input/docs/notes one.txt（内容为 alpha 和 beta 两行）、input/docs/.secret.txt（内容为 hidden）、input/tmp/empty.txt（空文件）以及 input/run.log（任意两行日志）。
- 显示 q01 的绝对路径，并以长格式列出 input 下全部项目，包含隐藏文件。
- 把 input 下所有.txt 文件复制到 work/<学号>/，保留相对目录结构并正确处理名称中的空格。
- 将复制后的目录权限设为 750、普通文件权限设为 640；生成 inventory.txt，列出相对路径和字节数。

(2) 解题过程

按任务要求输入指令如图

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ mkdir -p input/docs input/tmp

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo -e "alpha\nbeta" > "input/docs/notes one.txt"

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo "hidden" > input/docs/.secret.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ touch input/tmp/empty.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ echo -e "加油华为\n加油CHina" > input/run.log

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ pwd
/c/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/q01

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ ls -la input/
total 1
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:47 ./
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:42 ../
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:44 docs/
-rw-r--r-- 1 guolicheng 197121 25 Aug 29 17:47 run.log
drwxr-xr-x 1 guolicheng 197121 0 Aug 29 17:45 tmp/

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ mkdir -p work/25020007160

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find input -name "*.txt" -exec cp --parents {} work/25020007160/ \;
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find work/25020007160 -type d -exec chmod 750 {} \;

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find work/25020007160 -type f -exec chmod 640 {} \;

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ cd work/25020007160

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ find work/25020007160 -type f -printf "%p %s\n" > inventory.txt
find: 'work/25020007160': No such file or directory

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ find . -type f -printf "%p %s\n" > inventory.txt
```

(3) 实验结果

得到目录结构和 inventory.txt 内容如下，符合实验预期。

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01 (main)
$ find .
./input
./input/docs
./input/docs/.secret.txt
./input/docs/notes one.txt
./input/run.log
./input/tmp
./input/tmp/empty.txt
./work
./work/25020007160
./work/25020007160/input
./work/25020007160/input/docs
./work/25020007160/input/docs/.secret.txt
./work/25020007160/input/docs/notes one.txt
./work/25020007160/input/tmp
./work/25020007160/input/tmp/empty.txt
./work/25020007160/inventory.txt
```

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q01/work/25020007160 (main)
$ cat inventory.txt
./input/docs/.secret.txt 7
./input/docs/notes one.txt 11
./input/tmp/empty.txt 0
./inventory.txt 0
```

2. 题目二：随机访问日志统计

(1) 题目内容

在 q02 中创建 access.csv，并使用下方给定数据完成日志统计。

给定内容

```
timestamp,user,path,status,latency_ms
09:00:01,alice,/api/users,200,120
09:00:02,bob,/api/orders,500,310
09:00:03,alice,/api/users,503,180
09:00:04,carol,/login,500,90
09:00:05,bob,/api/orders,502,260
09:00:06,dave,/health,200,20
09:00:07,alice,/api/orders,500,420
09:00:08,carol,/api/users,500,150
```

题目要求

- 编写 analyze.sh，接收一个 CSV 文件路径；文件不存在时将错误写入 stderr 并返回非零退出码。
- 使用 Shell 管道以及 awk、sort、head 等工具，输出 HTTP 5xx 数量最多的前 2 个 path；按次数降序，次数相同时按 path 字典序。
- 输出全部数据行的平均 latency_ms，保留两位小数；表头不得计入。
- 分别对 access.csv 和一个不存在的文件运行脚本；不得使用 Python、电子表格或手工统计。

(2) 解题过程

按要求创建 access.csv，编写 access.sh 过程如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1 (main)
$ mkdir q02

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1 (main)
$ cd q02

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ vi access.csv

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ vi analyze.sh

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ chmod +x analyze.sh
```

```
# 脚本接收1个参数: csv文件路径
if [ $# -ne 1 ]
then
    echo "使用方式: $0 输入csv文件" >&2
    exit 1
fi

infile="$1"

# 判断普通文件是否存在
if [ ! -f "$infile" ]
then
    echo "错误: 文件 $infile 不存在" >&2
    exit 2
fi

echo "HTTP 5xx最多的前2个path:"
awk -F ',' 'NR>1 && $4 ~ /\^5[0-9][0-9]$/ {count[$3] = count[$3] + 1}
END {
    for(p in count) {
        print count[p], p
    }
}' "$infile" | sort -k1,1nr -k2,2 | head -2

echo ""
echo "latency_ms平均值:"
awk -F ',' 'NR>1 {total = total + $5; num = num + 1}
END {
    printf "%.2f\n", total / num
}' "$infile"
```

(3) 实验结果

分别对 access.csv 和一个不存在的文件运行脚本，结果符合预期

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ ./analyze.sh access.csv
HTTP 5xx最多的前2个path:
3 /api/orders
2 /api/users

latency_ms平均值:
172.22

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ ./analyze.sh 123.csv
错误: 文件 123.csv 不存在

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ echp $?
bash: echp: command not found

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q02 (main)
$ echo $?
127
```

3. 题目三：制造、解决并解释一次合并冲突

(1) 题目内容

在 q03 中从零创建一个 Git 仓库，并主动制造、解决一次合并冲突。

题目要求

- 初始化仓库，在 main 创建 config.txt，内容为 mode=normal，并完成初始提交。
- 创建 feature-a，把内容改为 mode=safe 并提交；从初始 main 创建 feature-b，把内容改为 mode=fast 并提交。
- 回到 main，先合并 feature-a，再合并 feature-b；解决冲突时保留 mode=safe，并另加一行 note=reviewed。
- 确认工作区干净，运行 git log --all --graph --decorate --oneline 查看历史。

(2) 解题过程

按题目要求初始化仓库和创建分支如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (main)
$ git init
Initialized empty Git repository in c:/Users/guolicheng/Desktop/系统开发工具基础/实验/week1/q03/.git/

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ echo "mode=normal" > "config.txt"

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git commit -m "mode=normal"
[master (root-commit) 85d1dc6] mode=normal
1 file changed, 1 insertion(+)
 create mode 100644 config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-a
Switched to a new branch 'feature-a'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ echo "mode=safe" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git commit -m "mode=safe"
[feature-a 64ed774] mode=safe
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-b
Switched to a new branch 'feature-b'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ echo "mode=fast" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git commit -m "mode=fast"
[feature-b 2efe8dc] mode=fast
1 file changed, 1 insertion(+), 1 deletion(-)
```

合并并产生冲突如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-a)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git checkout -b feature-b
Switched to a new branch 'feature-b'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ echo "mode=fast" > config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git add config.txt
warning: in the working copy of 'config.txt', LF will be replaced by CRLF the next time Git touches it

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git commit -m "mode=fast"
[feature-b 2efe8dc] mode=fast
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (feature-b)
$ git checkout master
Switched to branch 'master'

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git merge feature-a -m "合并feature-a"
Updating 85d1dc6..64ed774
Fast-forward (no commit created; -m option ignored)
 config.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git merge feature-b -m "合并feature-b"
Auto-merging config.txt
CONFLICT (content): Merge conflict in config.txt
Automatic merge failed; fix conflicts and then commit the result.
```

进入 config.txt 解决冲突如下

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ vi config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ git add config.txt

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master|MERGING)
$ git commit -m "解决冲突"
[master f10f4a2] 解决冲突
```

```
mode=safe
note=reviewed
```

(3) 实验结果

确认工作区干净后，查看历史如图，符合要求

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git status
On branch master
nothing to commit, working tree clean

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q03 (master)
$ git log --all --graph --decorate --oneline
* f10f4a2 (HEAD -> master) 解决冲突
|
| * 2efe8dc (feature-b) mode=fast
| * | 64ed774 (feature-a) mode=safe
|/
* 85d1dc6 mode=normal
```

4. 题目四：修复并构建一页技术说明

(1) 题目内容

将下方模板保存为 q04/report.tex，补全内容并从命令行构建 PDF。

给定内容

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{姓名-学号}
\maketitle
\section{Result}
% 在此加入公式、表格及正文引用
\end{document}
```

题目要求

- 加入一个带编号和 label 的公式，并在正文中使用 ref 或 eqref 引用。
- 加入一个 2 行 2 列表格，设置 caption 和 label，并在正文中引用该表。
- 使用 latexmk -pdf -halt-on-error 或 tectonic 生成 report.pdf。
- 确认 PDF 能打开，且交叉引用不显示 “??” 。

(2) 解题过程

按要求修改 report.tex 如图并生成 pdf

```
guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工具基础/实验/week1/q04 (main)
$ vi report.tex

guolicheng@LAPTOP-7D86CCBH MINGW64 ~/Desktop/系统开发工
具基础/实验/week1/q04 (main)
$ latexmk -pdf -halt-on-error report.tex
Initial win CP for (console input, console output, system): (CP936, CP936, CP936)
I changed them all to CP936
Rc files read:
  NONE
Latexmk: This is Latexmk, John Collins, 15 June 2025. ve
rsion 4.87.
No existing .aux file, so I'll make a simple one, and re
quire run of *latex.
```

```

\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{Zang Wenshang-25020007160}
\maketitle
\section{Result}

The arithmetic series summation is given by Eq.~\eqref{eq:series}.
\begin{equation}\label{eq:series}
\sum_{i=1}^n i = \frac{n(n+1)}{2}
\end{equation}

Experimental parameters are listed in Table~\ref{tab:param}.
\begin{table}[htbp]
\centering
\caption{Parameter Statistics}
\label{tab:param}
\begin{tabular}{|c|c|}
\hline
Sample Size & Valid Sample \\
\hline
200 & 186 \\
\hline
\end{tabular}
\end{table}

We can compute sequence sum using Eq.~\eqref{eq:series}. Table~\ref{tab:param} shows the experimental statistical parameters.
\end{document}

```

(3) 实验结果

经检验 pdf 符合预期，交叉引用未显示??

Tool Report

Zang Wenshang-25020007160

August 29, 2026

1 Result

The arithmetic series summation is given by Eq. (1).

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (1)$$

Experimental parameters are listed in Table 1.

Table 1: Parameter Statistics

Sample Size	Valid Sample
200	186

We can compute sequence sum using Eq. (1). Table 1 shows the experimental statistical parameters.

二、 课后练习

1. 题目一

(1) 题目内容

(2) 解题过程

(3) 实验结果

2. 题目二

(1) 题目内容

(2) 解题过程

(3) 实验结果

3. 题目三

(1) 题目内容

(2) 解题过程

(3) 实验结果

4. 题目四

(1) 题目内容

(2) 解题过程

(3) 实验结果

5. 题目五

(1) 题目内容

(2) 解题过程

(3) 实验结果

6. 题目六

(1) 题目内容

(2) 解题过程

(3) 实验结果

7. 题目七

(1) 题目内容

(2) 解题过程

(3) 实验结果

8. 题目八

(1) 题目内容

(2) 解题过程

(3) 实验结果

9. 题目九

(1) 题目内容

(2) 解题过程

(3) 实验结果

10. 题目十

(1) 题目内容

(2) 解题过程

(3) 实验结果

11. 题目十一

(1) 题目内容

(2) 解题过程

(3) 实验结果

12. 题目十二

(1) 题目内容

(2) 解题过程

(3) 实验结果

13. 题目十三

(1) 题目内容

(2) 解题过程

(3) 实验结果

14. 题目十四

(1) 题目内容

(2) 解题过程

(3) 实验结果

15. 题目十五

(1) 题目内容

(2) 解题过程

(3) 实验结果

三、 解题感悟

四、 GitHub 链接和 commit 记录截图

1. GitHub 链接

2. commit 记录截图